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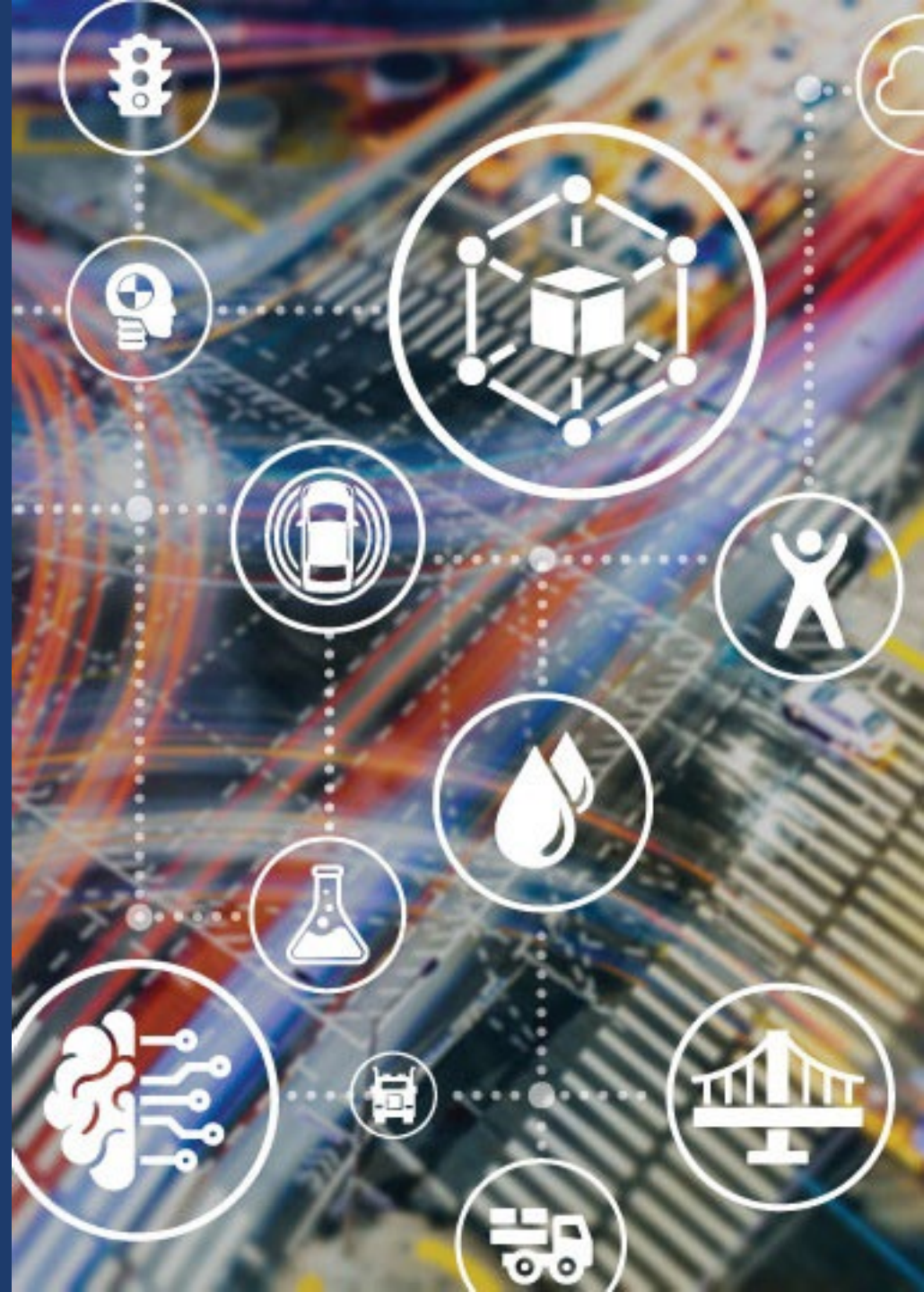
Traffic Analysis, Modeling, and Simulation (AMS)

Impacts of Connectivity and Vehicle Automation

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Agenda

- ▶ Introduction
- ▶ Connected and Automated Vehicle (CAV) AMS program and emerging trends
- ▶ Connecting the megatrends: Automation and connectivity
- ▶ Future effective capacity: The influence of different levels of automation
- ▶ Conclusions



FHWA Turner-Fairbank Highway Research Center (TFHRC)

- ▶ The U.S. Department of Transportation FHWA TFHRC is a federally owned and operated national research facility in McLean, VA.
- ▶ The center houses 15 R&D laboratories, including the Saxton Transportation Operations Laboratory.



Top: TFHRC aerial view. **Bottom:** Smart intersection.

All images source: FHWA.



CAV AMS Program

Vision: To support State/local agencies in understanding the most likely impacts of CAVs.

Goals:

1. Coordinate CAV AMS efforts across FHWA's disciplines.
2. Advance understanding of most likely CAV impacts on roadway performance.
3. Accelerate deployment of AMS tools and models considering CAVs.

Outcomes:

1. Facilitated active CAV AMS community.
2. Tracked CAV technologies and distribution.
3. Evaluated human behavior response to various types of CAV technology.
4. Validated system data for CAV AMS.
5. Made available regularly updated CAV AMS tool resources.



Emerging Technologies and Trends

Vehicle Automation:

- ▶ Trucking: Long-haul, middle-mile, last-mile delivery [L4].
- ▶ Shared mobility services: Ride hailing, transit (fixed route, on demand) [L4].
- ▶ Private vehicles: Widespread deployment of L1–L2 (advanced driver assistance systems (ADAS)); first approvals for Automated Driving Systems (ADS) L3 in United States in 2024.⁽²⁾

Connectivity:

- ▶ Traveler connectivity (navigation, transit, and 511 apps).
- ▶ Vehicle-to-everything (V2X).

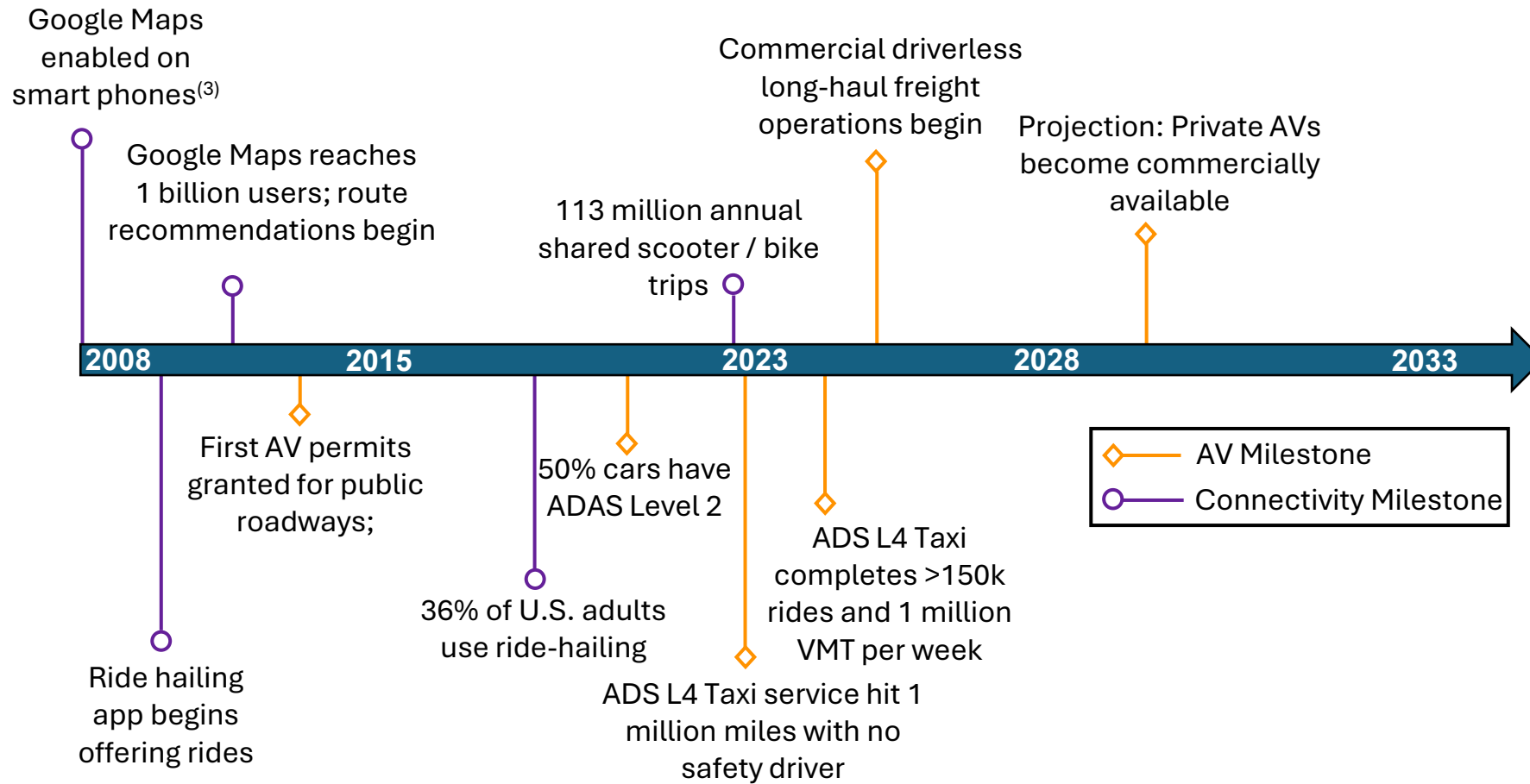
SAE® J3016™ Levels of Driving Automation

		SAE® Level 0™	SAE® Level 1™	SAE® Level 2™	SAE® Level 3™	SAE® Level 4™	SAE® Level 5™
What does the human in the driver's seat have to do?		You are driving whenever these driver support features are engaged—even if your feet are off the pedals and you are not steering.			You are not driving when these automated driving features are engaged—even if you are seated in the driver's seat.		
		You must constantly supervise these support features; you must steer, brake, or accelerate as needed to maintain safety.			When the feature requests, you must drive.	These automated driving features will not require you to take over driving.	
		These features are driver support features.			These features are automated driving features.		
What do these features do?		These features are limited to providing warnings and momentary assistance.	These features provide steering OR brake/acceleration support to the driver.	These features provide steering AND brake/acceleration support to the driver.	These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met.		This feature can drive the vehicle under all conditions.
	Example features.	• Automatic emergency braking. • Blind spot warning. • Lane departure warning.	• Lane centering. - OR • Adaptive cruise control.	• Lane centering. - AND • Adaptive cruise control at the same time.	• Traffic jam chauffer.	• Local driverless taxi. • Pedals and steering wheel may or may not be installed.	• Same as Level 4, but feature can drive everywhere in all conditions.

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The Accelerating Speed of Deployment



AV = automated vehicle; VMT = vehicle miles traveled.

Source: FHWA.



Impacts on AMS

How vehicle automation and connectivity influence AMS:

- ▶ **Changes in traffic patterns and behaviors:**

- ▷ Macrobehaviors: Route choice, departure time, parking, destination, etc.
- ▷ Microbehaviors: Car-following and lane changing.
- ▷ Longer term choices: Home locations, pickup/deliveries, long-distance travel.

- ▶ **New data sources and requirements:**

- ▷ Travel applications and crowdsourcing.
- ▷ AV data.
- ▷ V2X.

Important to understand and incorporate changing behaviors with AMS tools/methods to optimize traffic management and operational strategies in mixed-traffic environments!



White Paper Development

As part of the CAV AMS program, two white papers were developed to review literature and identify research/modeling gaps related to the following:

- ▶ Travel behavior changes and resulting traffic flow impacts from emerging technologies (AVs and traveler connectivity) are discussed in the first part of this presentation.⁽⁴⁾
- ▶ Highway capacity and throughput impacts from AV technologies (L1–L5), with a focus on how well theoretical studies align with real-world experiments, are discussed in the second part of this presentation.⁽⁵⁾





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Transportation Megatrends



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Focus of Literature Review

1. Travel behavior changes and new modeling approaches considering AVs and traveler connectivity.
2. Potential synergies and conflicts between the megatrend technologies.
3. Review of AMS tool capabilities related to modeling changing travel/driving behaviors from emerging technologies.
4. Identification of future research topic areas based on identified gaps/challenges.



Key Findings: Behavioral Changes Affecting AMS

▶ **Traveler connectivity:**

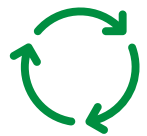
- ▶ Digital navigation and recommendation systems provide new opportunities for individual travelers to make selfish travel choices, sometimes leading to misuse of infrastructure (e.g., using local roads for cut-through traffic).^(6,7,8)
- ▶ Shared service models are affecting traffic operations in new ways, which are not always captured in AMS tools (fleet-optimal routing, dead-heading, in-lane pickup/drop-off).^(9,10)

▶ **AVs:**

- ▶ Lower value of travel time alters travel choices and AMS assumptions, such as long-distance travel modes, home locations, departure times, and route choice.^(11,12,13)
- ▶ Zero-occupancy travel behaviors differ from human drivers (e.g., congestion seeking, increased VMT to reduce parking costs).⁽¹⁴⁾



Traveler Connectivity Plus AVs



Synergies

► Network efficiency:

- ▷ Real-time information and automated decisionmaking can improve system performance under a system-optimal policy framework.
- ▷ Improved representation of surrounding conditions that augment data received from onboard sensors further improve decisionmaking tasks.



Conflicts

► Network inefficiency:

Without regulation, AVs will be programed to maximize individual benefits at the expense of the greater network. Rational (selfish) route choice could result in large variations in travel time on major parallel routes.

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Traveler Connectivity Plus AVs (continued)

Synergies

- ▶ **Affordability (shared model):**
Removing labor costs from shared service models reduces overall operational costs, which can then be passed to travelers.

Conflicts

- ▶ **Affordability (private ownership):**
Advanced automation and connectivity come at an initial premium but offer much lower travel costs, attracting more use of low-occupancy vehicles. This expanded use may increase congestion as well as draw customers from high-occupancy vehicles (transit).

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Focus Areas for AMS Enhancements

Network utilization:

New behaviors from real-time information and recommendations, which require more detailed network representations that include bridge clearances, steep grades, sharp turns, and lane widths, among others.

Zero-occupancy trips:

- ▶ New objective cost functions with higher weight given to monetary costs.
- ▶ Tradeoffs between access time (for private owner pickup/drop-off) and parking costs.

Shared service models:

- ▶ Fleet behaviors versus individual travelers.
- ▶ Interactions at the curb (pickup/drop-off, circling).



Considerations for Future Research (1/2)

- ▶ Study and differentiate travel behaviors and risk attitudes by vehicle technology type.
- ▶ Develop improved understanding of human responses to real-time recommendations and resulting impacts to network efficiency.
- ▶ Identify changing behavioral patterns from hands-free features (SAE Level 2/Level 3).⁽¹⁾ For example, will travelers choose longer routes to enable hands-free features on specific freeways?
- ▶ Develop new cost functions for zero-occupancy travel and decisionmaking—private SAE Level 4/Level 5.⁽¹⁾
- ▶ Integrate fleet-optimal algorithms and other fleet behaviors (in-lane pickup/drop-offs, deadheading)—shared SAE Level 4/Level 5.⁽¹⁾





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Future Effective Capacity

*Assessing Traffic Flow Impacts for Variable Levels of Automation
and Market Acceptance*



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Future Effective Capacity White Paper

► Purpose:

- ▷ Review and summarize literature evaluating AV impacts on freeway capacity and throughput in mixed-traffic environments (SAE Level 1 to Level 5).⁽¹⁾
- ▷ Summarize how theoretical car-following and lane-changing models are being adapted to best capture these new AV behaviors (inputs, assumptions, etc.).
- ▷ Compare theoretical assumptions with real-world experiments to identify inconsistencies.
- ▷ Identify future research, data-collection, and deployment activities to fill research gaps related to modeling AVs and human drivers in realistic, mixed-traffic environments.

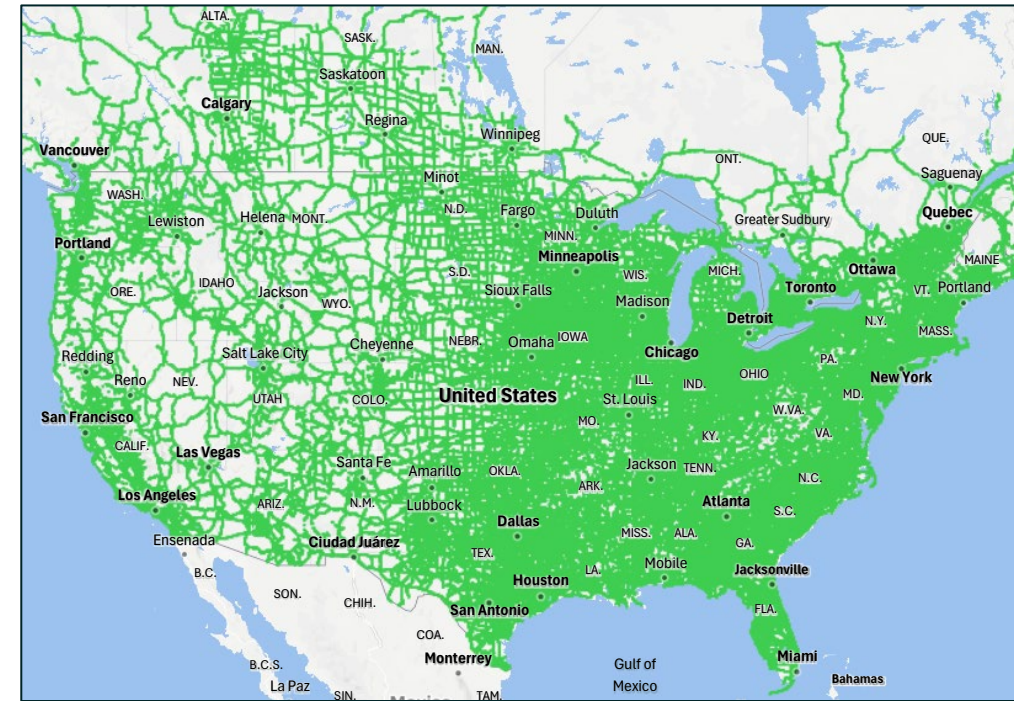
► Scope:

- ▷ SAE Level 1 to Level 5 vehicles.⁽¹⁾
- ▷ Microsimulation applications: Car-following and lane-changing behaviors.
- ▷ Human driver response.



Recent Trends

- ▶ **L2+ (hands-free, constant supervision on selected roads)**
- ▶ **L3 (hands-free, eyes-off in specific operational design domains)**
- ▶ **L4 (driverless operations)**



© Mapbox.⁽¹⁵⁾

Example of a L2+ Network



Future Projections: Automation Mix⁽¹⁶⁾

BAS ^t level	Year	2015	2020	2025	2030	2035	...
ACC		2%	5%	10%	15%	20%	
ACC + LCA			<1%	2%	10%	20%	
High Automation				<1%	3%	6%	
Full Automation							???

ACC = adaptive cruise control; BAS^t = Federal Highway Research Institute (German); LCA = lane-change assist.



Currently Available Longitudinal/Lateral Controls

Less
Automation



More
Automation

Longitudinal Controls:

- ▶ Warnings, momentary assistance (e.g., automatic emergency braking, forward collision warning).
- ▶ Automated control (e.g., ACC, cooperative adaptive cruise control).

Lateral Controls:

- ▶ Warnings, momentary assistance (e.g., lane departure warning, LCA).
- ▶ Lane keeping and lane centering (e.g., lane-keeping assist, lane centering).
- ▶ Automatic lane changing—performs lane change when driver desires.
- ▶ Autonomous lane changing—automated system makes decision and executes lane change.



Key Findings: Unrealistic Modeling Assumptions

1. **Theoretical studies assume perfect technology performance and limited impacts from human factors:** e.g., real-world testing of commercial ACC has shown longer reaction times, longer headways, and greater instability compared to human drivers.^(13,16)
2. **Automated lane-changing behaviors and human responses are rarely considered.**^(13,17)
3. **Homogenous AV behavioral assumptions are not realistic.** Automated features have different capabilities in different environments and vary by manufacturer.⁽¹⁸⁾
4. **Models lack analysis in real-world environments, use imperfect automation, and do not consider various vehicle types.** Automated feature controls differ across weather conditions, road grades, complex configurations, vehicle size/type, etc., which will impact automated system behaviors, human responses, and ultimately, throughput.⁽¹⁸⁾



Key Findings: Limited Understanding of Human-Machine Interactions

1. **Lack of research related to human–machine interactions and cooperation when using L1–L3 features.**

Capacity and throughput impacts of takeover events, transfer of control, and failover procedures are not well understood.⁽¹³⁾

2. **Limited research related to changing human driving behaviors when interacting with automated vehicles.**

Do drivers behave more aggressively or conservatively when interacting with AVs? How does this behavior impact drivers' car-following and lane-changing behaviors? ⁽¹³⁾



Key Findings: Current AMS Frameworks Struggle to Capture Realistic AV Behaviors

1. **Difficult to modify current models to accommodate realistic AV behaviors.**

Need to develop new frameworks that capture the underlying AV decisionmaking process that includes sensor inputs, data fusion, recognition, machine learning, and optimized decisionmaking.^(13,16)

2. **Idealistic assumptions about AV feature capabilities.**

Controllers vary based on vehicle type, manufacturer, and roadway conditions, among other factors. Simple assumptions about reduced headways and faster reaction times are not realistic.^(13,16)

3. **Limited understanding of AV lane-changing behaviors.**

Trajectories, risk attitudes, execution.^(13,17)

4. **Lack of studies assessing impacts to capacity/throughput due to improved safety from Avs.**

Smoother trajectories, reduced speed variance, etc.⁽¹⁶⁾



Considerations for Future Research (2/2)

- ▶ Evaluation of currently available automated features' (L2+, L3) car-following and lane-changing behaviors across original equipment manufacturers and conditions.
- ▶ Human responses to automated features at various levels (L1–L5) and in different operating environments.
- ▶ Impacts of human–machine interactions/cooperation on traffic flow and stability (takeover events, failover protocols, etc.)
- ▶ Development of new traffic theories to better capture the AV decisionmaking processes.
- ▶ Quantification of traffic flow impacts due to changes in the number of incidents/collisions that might occur in a mixed-traffic environment.



Conclusions

- ▶ Traffic systems for the next several decades will have a mix of L1–L5 technologies and human drivers, which need to be captured in AMS tools for near- and midterm planning, design, and investments.
- ▶ Emerging technologies are changing travel/driving behaviors, which need to be captured with AMS tools for accurate results.
- ▶ A need exists to capture realistic automated feature behaviors (reaction times, headways, etc.) and incorporate them into AMS for near-term planning.
- ▶ A lack of understanding exists about how automated features behave in various scenarios/situations and how humans respond to these behaviors.
- ▶ Thoughtful and strategic integration of emerging technologies is needed to maximize system-level benefits in the near and long term.



Questions?



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